

THEORY • HYPOTHESES • DIRECTIONS FOR INVESTIGATION

FROM RULES, WORLDS

Existence, prediction and equivalence
under finite resources

A research program
for humans and AI systems

FOUNDING INTUITION

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Analytical development in collaboration
with an AI partner

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Two worlds that seem identical

There is a mechanism in a room. When we press a button, a ball moves forward, strikes a flap, and the flap opens a gate. In the next room there is a storyteller. When we press the same button, he describes a ball moving forward, striking a flap and opening a gate. If our access to the two rooms consists only of a sequence of sentences, the initial accounts may be identical. Yet we have the impression that something happens in the first room that is merely spoken in the second.

The impression deserves to be taken seriously, but it is not yet a definition. The storyteller may faithfully follow an imaginary mechanism. The mechanism may be connected to a hidden device that changes its behavior. Moreover, the storyteller himself is a physical system. The production of a description and the world described are not the same object of analysis. The distinction we are trying to capture cannot be reduced to an opposition between matter and words.

Let us continue the scene. We block the ball before the flap and ask whether the gate still opens. We change the distance between the parts. We ask for the consequences of an intervention that has never been encountered before. A storyteller may answer correctly, and a faulty mechanism may respond surprisingly. What begins to emerge is not a magical test of existence, but a program for investigating dependencies: what must be preserved across situations for an outcome to be explained and anticipated?

The book starts from an intuition attributed to the human author, Sînică Alboaiie: reality seems to bear the obligation of continuity of consequences, whereas a representation may introduce any number of shortcuts, new rules or exceptions. Perhaps understanding is precisely the construction of a representation in which these shortcuts become justified. Perhaps the difference between an explanation and an improvisation is not the number of rules, but the way the rules remain tied to what they can produce, what they forbid and what it costs to apply them.

This idea is not presented as a proven discovery about the universe. It is the starting point of a program. The program should be able to survive even if the world has no short fundamental description. It should remain useful if some phenomena are irreducible at the level of trajectories, if observers see only fragments and if there are several good models rather than just one.

The ambition is to understand when a long history can be replaced by a small state, when many regularities can turn into a common procedure, and when a simplification destroys precisely what we needed. The practical objective is to construct representations that can be used, checked, corrected and transferred between tasks. The philosophical objective is more cautious: to discover what entitles us to call a regularity a law, a compression understanding, and an equivalence something more than a convenient resemblance.

Intuition, analysis and responsibility

The founding intuition and philosophical orientation belong to the human author. The mathematical exposition, distinctions and methodological proposals are developed with AI assistance. This attribution does not mean that every technical hypothesis represents a conviction already endorsed by the human author. When the text introduces a working definition or a research objective, these should be read as proposals open to evaluation.

The ‘AI partner’s assessment’ means an argumentative position: which formulations are coherent, which consequences follow from the premises, which confusions must be avoided, and which directions seem productive. An AI system’s agreement is not independent verification and does not constitute scientific consensus. It does not replace proof, the scholarly literature or confrontation with new observations.

The book separates four registers. A philosophical commitment offers a perspective on the world without automatically being measurable. A technical definition establishes a common language. A mathematical proposition holds under its stated assumptions. An empirical hypothesis anticipates a phenomenon that could be observed or refuted. Constructed examples are tools for thought, and the proposed investigations are prospective. They do not represent empirical results already obtained.

The numbered references indicate the existing foundations. The intended contribution is not to claim these foundations, but to articulate a common question and a working method that combines them. The bibliography is selective: it offers entry points for a program and does not claim to exhaust the fields it crosses.

From intuition to investigation

The first part reconstructs the intuition in its strongest form and distinguishes it from claims it does not justify. The second part introduces systems, observers, costs and equivalence relations. The third part examines sequential prediction, learning operations and the relationship between language and world modeling. The fourth part describes families of investigations that could yield interpretable results. The final part formulates hypotheses, attainable mathematical results, rules of collaboration and questions that remain open.

Readers interested in philosophy can follow the first four chapters and return to the conclusions after the chapters on causality and mind. Technical readers particularly need the definitions in chapters 5–10 and the protocols in chapters 14–18. An AI system used as a research partner should also read the chapter on epistemic discipline: the most costly failure of such a program would be the premature transformation of a fruitful idea into a self-confirming vocabulary.

Mathematical concepts are introduced after the problem they solve. Accepting a computationalist metaphysics is not necessary to use the tools. Preserving one distinction is necessary, however: a model can be sufficient for a question without being a complete description of the thing we are asking about.

1. The intuition of a world that does not improvise

1.1. The obligation to continue

The human author's intuition begins not with a formula, but with an asymmetry. In a mechanism we construct, a consequence must occur through existing parts and permitted interactions. In a story, a consequence can be introduced by a single sentence. We require the mechanism to produce the result; for the sentence to exist as a sentence, stating it is enough.

This does not mean that every sentence is arbitrary or that every mechanism is intelligible to us. It means there are two different obligations. One is the obligation of realization: how does the system move from one state to another? The other is the obligation of description: what does the statement claim is happening? A good explanation connects them. An explanation that merely renames the result leaves them separate.

Suppose a box delivers the correct answer to a question. We can describe the operation through the rule 'the box solves the problem'. The description is short, but it has hidden precisely the difficulty. The human author suspects that an important part of understanding consists in disallowing such unpaid shortcuts: we must know what information enters the box, what transformation takes place, and under what conditions the answer remains correct.

On this reading, 'tracking results' does not presuppose a cosmic memory of all events. It expresses the fact that the past cannot be erased for free when it still influences the future. An object moved to another room cannot reappear in the first merely because the narrator forgot the move. The model must either preserve the relevant information or introduce an explicit mechanism for returning it.

1.2. Why a few principles seem sufficient

When the same operation applies in different situations, observations become organized. We no longer have a collection of happenings, but a way to generate consequences. A conservation rule, a transition relation or a composition of operations can replace a long enumeration. The intellectual attraction of a general rule comes from this economy, but also from the new obligations the rule creates: it excludes continuations that previously seemed possible.

The human author's intuition is that the world might be rich primarily through the unfolding of rules rather than their arbitrary multiplication. This possibility is subtler than the idea that 'everything is simple'. A compact source can produce objects that are difficult to compute, recognize or compress into a form useful to an observer. Simplicity of origin and simplicity of access to consequences are different things.

That is why the promising meaning of the intuition is not a promise of easy explanation. It is an invitation to search for reusable structures. Instead of assuming a separate law for every phenomenon, we can ask whether two phenomena are unfoldings of the same transformation in different states. When this strategy succeeds, knowledge gains the power of transfer.

But local success does not automatically justify a universal conclusion. A cautious formulation would be: in many well-defined problems, it is worth searching for a mechanism more compact than an exhaustive description of behavior. A much stronger formulation would be: all reality must have a finite algorithmic basis. The research program needs the first as a heuristic; it need not assume the second as a truth.

1.3. A leap can be a discovery, not a fraud

A shortcut is not necessarily an error. If we know that repeating an operation of a certain type is equivalent to a direct expression, we can replace a long unfolding with a short calculation. This is often precisely the value of a theorem, a suitable representation or a compiled operation.

The difference is justification. ‘This is the result because I declare it’ and ‘this is the result because the transformation preserves the relevant property’ may have the same verbal length, but they do not have the same epistemic structure. The second formulation incurs a debt of proof or validation. That debt can be paid exactly, approximately or only within a restricted domain.

The human author sees a possible bridge here between imagination and reasoning. Imagination proposes leaps; reasoning tries to connect them to consequences. A library of mental operations could grow without the world acquiring new fundamental laws. The vocabulary through which the agent navigates the world grows, and the cost of certain questions decreases.

This is a functional hypothesis, not a neurobiological claim. It allows us to investigate artificial systems that invent representations and ask whether the new representations actually shorten the solution of unseen problems. Giving a procedure a new name is not enough. It must achieve savings that survive accounting for memory, learning and verification.

1.4. The core worth preserving

The intuition can be condensed as follows: between the unfolding of a process and the representation of its consequences lies a distance that can be reduced, but not always and not for free. Sometimes a new representation discovers real simplicity for the question being asked. At other times it creates only an impression of simplicity by hiding information or computation.

The proposed program addresses this distance. It does not demand an absolute separation between real and imaginary worlds. It asks us to explain what the coherent continuation of a world means, how much of the past must be preserved, what kinds of shortcuts are legitimate, and how to detect the point at which a representation ceases to support the consequences it asserts.

The founding question is therefore not only ‘how many rules is the world made of?’ but also ‘from what kind of representation can we correctly continue the world at a cost we can afford?’ The first ques-

tion remains philosophical and physical. The second opens common ground for information theory, programming-language semantics, learning and AI system design.

2. Existence, reduction and laws: what we actually assume

2.1. Four meanings that must not be mixed

An object can exist physically, a structure can exist in a mathematical sense, a mechanism can be realized in a technical system, and a character can exist as an object of discourse. These expressions do not serve the same function. A mathematical existence proof does not automatically produce a physical realization. A coherent description of a character does not prove its existence outside the description. A technical process, by contrast, is a physical process even when it represents a fictional world.

The book does not propose a universal arbiter among all meanings of existence. It introduces a working boundary: it will study systems described through states, transformations and observations, along with agents that construct representations of these systems. This boundary does not assert that every aspect of existence is exhausted by this form. It asserts that an important class of questions about prediction can be formulated this way.

The difference between consistency and realizability also matters. A description without obvious contradictions may remain incomplete. Some mathematical theories describe objects without providing a finite procedure for constructing them. Some technical scenarios assume unlimited resources. Consistency, constructibility and physical possibility must be evaluated separately; otherwise a single expression, ‘exists’, takes on tasks it cannot fulfill.

2.2. Reductionism does not mean merely a few rules

Reductionism can designate an ontological commitment: phenomena depend on processes at a more fundamental level. It can also designate an explanatory strategy: we understand a phenomenon through

its components and interactions. A third, stronger claim would be that a high-level explanation can be efficiently reconstructed from a low-level description. The first two do not automatically imply the third.

Anderson formulated an influential critique of identifying knowledge of fundamental laws with the ability to reconstruct all complex phenomena. The relevant idea here is the autonomy of problems of organization at different scales, not necessarily the denial of physical dependence. [2]

For our program, this distinction is decisive. A high-level state can be more useful than a complete list of components. ‘The gate is blocked’ may be sufficient for planning without reproducing every microscopic movement. Going deeper does not always mean explaining the particular question better. But neither does the usefulness of a high-level description demonstrate the emergence of additional fundamental laws.

Thus, one can accept dependence on a substrate without assuming that all explanations must be expressed in the substrate’s vocabulary. The program favors a disciplined pluralism of representations: multiple levels, linked through explicit correspondences, with known domains of validity and costs.

2.3. Law as mechanism or summary

The philosophy of laws of nature offers different positions. Some treat laws as systematized regularities; others connect them with necessity or the powers of things. ‘Best system’ analyses give a role to the balance between the simplicity of a description and its explanatory strength. These positions are not equivalent, and no single technical criterion can decide between them. [1]

The distinction has a methodological consequence. If a law is a good summary, improving compression can count directly toward its status. If a law is a real constraint on what is possible, compressing observations is only an indirect route to knowing it. The same equation can be used by researchers with different philosophical commitments.

In this book we accept a rule of caution: we call any transformation declared in a representation a ‘model rule’. We call a regularity that survives relevant confrontations with new situations a ‘candidate law’. We do not automatically turn descriptive success into evidence of metaphysical necessity. This reservation does not weaken research; it prevents an unannounced change of question.

2.4. What cannot be deduced from finite observations

Imagine two sources that emit the same sequence for the first thousand moments. The first continues forever according to a simple rule. The second is defined to coincide with the first until moment one thousand, then behave differently. No observation limited to that interval can distinguish them. The construction does not prove that all models are equally good; it proves that reliably identifying a global law requires additional assumptions.

These assumptions may concern the class of mechanisms, stability, a preference for compact descriptions or access to interventions. They must not be hidden. When we say a rule has been ‘discovered’, we should specify against which alternatives and with what type of data it was favored.

There is also a distinction between an unknown deterministic process and a process with intrinsic randomness. For a limited observer, both may induce probability distributions. Using probabilities does not mean deciding the metaphysics of chance. It means, first of all, correctly representing the available predictive uncertainty.

2.5. A realism sufficient for work

The program needs a minimal commitment: results must not change merely because the researcher prefers a different conclusion. In an artificial system, this independence is achieved through a specification and separate evaluation. In relation to the physical world, it is sought through measurement, replication and controlled interventions.

We do not need to prove that the universe is a computer to compare two models of a mechanism. Nor do we need to settle the status of mathematical objects to prove that two finite representations preserve the same property. The advantage of this boundary is that it allows researchers with different metaphysics to cooperate on common, testable questions.

3. When the rules seem to change

3.1. The same observation can hide different mechanisms

A system that usually repeats the last symbol may, after a while, begin to invert it. We can say that ‘the rule has changed’. But the expression covers different explanations. Perhaps there is a hidden switch. Perhaps a parameter varies continuously. Perhaps the external environment intervened. Perhaps our initial model omitted part of the state.

A library that receives a new operation exhibits another kind of change. The operation may be a composition of the old ones. In that case the repertoire available at a particular level changes, but not necessarily the fundamental computational power. If we introduce a new physical resource or external access to information, the change is different and must be recorded as such.

The program therefore distinguishes changes in parameters, selection of a regime, growth of memory, extension of the operational vocabulary and modification of the interface with the environment. Aggregating them into a single counter of ‘new rules’ would hide precisely the mechanisms we want to understand.

3.2. Absorbing rules into state

Consider a state x and a repertoire of rules r . The dynamics are given by two transformations: one updates the state using the repertoire, and the other updates the repertoire using the state. For discrete time, we write:

$$x(t+1) = F(x(t), r(t))$$

$$r(t+1) = U(x(t), r(t))$$

If both transformations are effective, we can define the extended state $z(t)$ as the pair consisting of $x(t)$ and $r(t)$. Then there is a fixed transformation G that updates the pair. We have not proved that the system is simple. We have shown that changes of rules at one level can be represented as changes of state at another.

This observation has an important condition. The description of the update must be available and must actually cover the relevant changes. If r receives unknown external information, it must be modeled as inputs or exogenous variables. If we include the entire history of the universe in the initial state, we obtain a formally fixed description that may be useless and enormous.

Moreover, the size of an extended state can grow without a fixed bound over time. A fixed interpreter working with growing memory is not a finite automaton. Confusing ‘fixed rule’ with ‘fixed memory’ would destroy the analysis of learning we wish to build.

3.3. Relative novelty and fundamental novelty

Suppose an agent learns to recognize a configuration and directly execute a long sequence of steps. From the agent’s perspective, a new operation appears. From the perspective of the underlying system, only a new memory object may appear. Both descriptions can be correct because they refer to different levels.

Relative novelty should not be minimized. A new concept can radically transform which problems are accessible. The difference between having a procedure available and having to rediscover it at every use is real for a finite agent. But this difference must be expressed through access, cost, transfer and stability, not through an improvised metaphysics of the emergence of laws.

Fundamental novelty would require a stronger claim: the new transformations cannot be represented within the previous framework by the permitted means. Such a conclusion is always relative to the stated framework or requires a much broader physical and mathematical argument. Observing a surprise is not enough.

3.4. Randomness does not invent meaning on its own

A probabilistic mechanism can produce new sequences without changing the distribution from which it draws them. The number of possible realizations may be very large, and each realization may contain new information for the observer. This is not equivalent to inventing a new law.

If an agent draws a random expression and adds it to a library, syntactic novelty is easy to obtain. Semantic value arises only if the expression has relevant, testable effects. A new name, a new string and a new capability are three distinct phenomena. A research program on creativity must measure them separately.

Likewise, the appearance of a rule stated in a text does not yet tell us whether it has been integrated into the internal model. The relevant test is whether the system preserves its consequences in new situations and can detect incompatibilities with other commitments. A sentence can announce a change that subsequent behavior fails to respect.

3.5. What discovering the metarule would mean

The existence of a representation with fixed dynamics does not automatically provide an algorithm to find it from observations. This is an identification problem. From limited data, a model with a hidden switch and one with successive exceptions may make similar predictions. A good investigation must create situations in which the explanations diverge.

Sometimes observing more will suffice. At other times we must intervene on a factor. At still other times the identity of the metarule remains undetermined, but we can find a predictive state that works. The program must accept this last result as a limited success rather than turn it into a discovery of the real mechanism.

4. What the AI partner accepts and where agreement ends

4.1. An assessment, not a vote of authority

The AI partner's assessment favors the intuition as a research program, but is conditional regarding its ontological formulation. Behind the contrast between few and many rules lies a precise problem: the same family of behaviors may have descriptions with very different memory requirements, costs and transfer capabilities. Identifying the relationships between these descriptions is a real technical target.

The AI partner also accepts the distinction between stating a consequence and supporting it through a mechanism. This helps separate fluency from fidelity and a useful representation from a merely plausible explanation. However, the mechanism supporting the answer can be statistical, symbolic or hybrid. There is no reason to reserve fidelity for a single type of implementation.

Agreement also concerns the value of studying approximations. A model does not need to copy every detail of the world to be useful. But we must know what it omits and why the omission does not change the relevant answers. Here lies the possibility of a solid link between philosophy and engineering.

4.2. What needs reformulation

The first reformulation concerns the number of rules. This depends on the vocabulary. A complex instruction can hide an entire library, while a library can be decomposed into many elementary instructions. The working measure must include the complete description of the mechanism under the chosen convention, not merely the number of visible labels.

The second reformulation concerns ‘existence in words’. A text is a physical or digital object, and its production has a mechanism. The world to which the text refers may be real, fictional, incomplete or contradictory. The analysis must distinguish the process producing the text from the process represented by it.

The third reformulation concerns irreducibility. An evolution that is difficult to accelerate is not automatically devoid of exact equivalences. Two systems can do the same work in different forms. Conversely, a fast approximation may preserve an aggregate property without preserving the trajectory. Equivalence, acceleration and discovering an equivalence are separate problems.

4.3. What is not accepted as a premise

The AI partner does not treat it as a fact that reality must have few fundamental rules. It does not treat next-symbol prediction as a complete theory of mind. It does not assume that a probabilistic mechanism is less real or that a symbolic representation is inherently more explainable. Nor does it assume that every useful regularity can be discovered efficiently.

These reservations are not rejections of research. They protect precisely what is most interesting: the possibility of finding out in which classes of systems, at which scales and under which resources better representations exist. If we fixed the answer in the premises, research would become merely a search for favorable examples.

Nor is reducing all problems to a single objective function accepted without justification. A predictive average may favor ignoring rare events. A compression measure may penalize a distinction essential for interventions. A short response time may hide an enormous preparation cost. The criteria must remain sufficiently separate to make these tradeoffs visible.

4.4. The positive wager

The wager the AI partner considers valuable is this: part of learning can be understood as discovering the relevant state, and part of expertise as turning costly unfoldings into reusable operations with conditions of validity. Not all learning reduces to these, but they are components that can be studied in isolation.

The program could produce a map of situations in which an agent must increase memory, situations in which it can compress, and situations in which it must change representation. It could show that some failures attributed to insufficient data are actually insufficiencies of state. It could also show the opposite: that an elegant representation cannot overcome a lack of relevant information.

This position does not turn the AI partner into a guarantor of human intuition. It obliges it to expose that intuition to the risk of losing. An idea that can explain any result, including the opposite of what was anticipated, does not yet offer a scientific direction. The direction becomes real when we can describe a result that would force us to change the model and know which part of the intuition would be affected.

5. The system, the state and the observer

5.1. What a generative description must contain

A generative description is not merely a sentence summarizing what happens. It must allow a continuation, or a distribution of continuations, to be determined. To begin, we can work with discrete systems observed at successive moments. This choice simplifies the analysis; it does not assert that physical space and time are discrete.

We will call S the set of possible states, A the set of actions available to the observer, and O the set of observations. A transition kernel K specifies the probability of the next state, conditional on the present state and action. An observation channel E specifies what the observer can see. The distribution μ describes the initial conditions. Together, these components form the working model:

$$M = (S, A, O, K, E, \mu)$$

In a deterministic mechanism, K places all probability on a single continuation. In a system without interventions, A may contain a single action, ‘let the system evolve’. The observation can be the complete state or only an indicator. Thus, the same language can describe an automaton, a noisy process or an agent experimenting with an environment.

A simple example is a tank. The state includes the water level and the position of a valve. The action may open or close the supply. The observation may be only a light that comes on above a threshold. Two different levels produce the same observation, but not necessarily the same future. The light is a sufficient description for the question ‘is the threshold exceeded now?’ but insufficient for ‘how much time remains until overflow?’

5.2. State is not synonymous with the entire past

We call a representation predictively sufficient if it preserves the information from the past needed for the continuations of interest. If we knew the valve position and the water level, we might no longer need all previous measurements. If we do not observe the valve, the history of levels might help us infer its position.

Formally, we can make almost any process dependent on the past ‘dependent only on the present state’ by defining the state as the entire past. This transformation is correct, but it does not solve the memory problem. A useful research program must distinguish the existence of a sufficient state from the existence of a sufficient state that is compact and easy to update.

There is another often overlooked requirement. It is not enough for a representation to provide the right answer at one moment. It must be updatable after the next observation. If updating again requires the entire past, the apparent compression has moved the cost into a hidden stage. The operational representation must include both the answer and a continuation rule.

5.3. The observer is part of the question

Two models can appear identical to one sensor and different to another. An observer who sees only average temperature does not distinguish every microscopic configuration. An observer tracking a particular particle requires different information. This does not mean truth is arbitrary, but that equivalence is relative to the declared access and questions.

The observation channel must not be chosen after seeing the result in order to hide differences. If all states are transformed into the same symbol, any two systems become observationally identical. This is a formal equivalence, but it has no value for the original task. Relevant observations are therefore fixed before comparison, and the loss of any observations must be justified by the research purpose.

The same discipline applies to time. One model may jump directly to a final result while another presents every stage. They may be equivalent for the final question and nonequivalent for continuous monitoring. The unit of time, observation frequency and what must be emitted between input and output are part of the contract.

5.4. Technical assumptions that are not cosmological beliefs

For the initial investigations we accept effectively executable descriptions, well-defined observation sets and controlled access to data. If we use probabilities, we specify how they are estimated or supplied. If we assume stationarity, we declare that the statistical mechanism does not change over the relevant interval. If it does change, we explicitly include the clock, regime or process of change.

These assumptions do not claim that the universe is a finite computer, that all physical variables are observable, or that any distribution can be learned. They define a domain in which the questions have precise meaning. The domain must be extended in stages: continuous observations, finite precision, unbounded memory, uncontrolled environments and partial access to interventions.

At the engineering level, we accept only comparisons in which data and resources are inventoried. A model receiving the hidden state solves a different problem from one seeing only the text. Both can be studied, but they cannot be declared fair rivals without acknowledging this difference. Even the choice of symbols can communicate part of the solution.

5.5. The first unit of research

The basic unit will not be ‘a world’, but a complete situation: a generator, an observation channel, a set of interventions, a family of questions and a budget. This choice may seem modest compared with the ontological ambition, but it allows knowledge to accumulate. We can modify a single element and observe what changes.

For example, we keep the generator and degrade the sensor. If the model needs more memory, we have identified a cost of incomplete observation, not an increase in fundamental complexity. Or we keep the sensor and extend the horizon. If new distinctions arise, we have identified a cost of the question. Separating these effects is already a form of theoretical progress.

6. There is no single number called complexity

6.1. Rule length and hidden information

Let us describe a world through the instruction ‘display the next entry in the table’. The instruction is very short. If the table contains a billion independent results, the complete description is not short. Information has been moved from rules into data. The same can happen with a numerical constant: a single value, required at enormous precision, can hide an enormous amount of information.

A conceptual accounting must therefore include the mechanism, parameters, initial state and exogenous information. For a probabilistic model, the description of observations can be divided between the length of the model and the cost of encoding the data under its distribution. This is the basic idea of the minimum description length principle, which provides precise statistical methods, not merely an aesthetic preference for simplicity. [3, 4]

$$L(\text{total}) = L(\text{model}) + L(\text{data} \mid \text{model})$$

The formula does not claim that there is a single natural encoding. The description convention, parameter precision and model class must be specified. In finite comparisons, these choices can change the result. A candidate must not be favored by a special instruction that compresses its exact solution without accounting for the instruction’s definition.

6.2. A short probabilistic law can generate long histories

The rule ‘independently produce zero or one with equal probabilities’ is compact. Yet a particular long realization may have no description much shorter than itself. The law describes the distribution, not all individual outcomes. Confusing the two would make any random source appear to contradict generative simplicity.

For a deterministic system using an external random tape, the tape must be included when we require exact reproduction of the trajectory. If we require only reproduction of the distribution, correctly specifying the source of randomness is sufficient. These are two different tasks with two different accountings.

This distinction limits a tempting ontological conclusion. Even if the laws of nature had a short formulation, it would not follow that all particular facts could be compressed to the same length. In such a situation, the law would constrain possibilities without choosing every realization by itself.

6.3. Memory, time and the difficulty of finding the model

A model with two logical states can have a large implementation. A model with many states can use a highly regular representation. The number of states, occupied memory and entropy of the distribution over states are not the same measure. If states are visited very unevenly, the average information needed to identify them may be small, although storing the complete structure remains costly.

Time must also be divided. There is the time needed to discover the model, the time needed to prepare a query, and the response time. A representation that is difficult to learn may be advantageous if reused enough. A representation that pays off only after an unrealistic number of uses is not automatically a relevant engineering achievement.

Short description and long running time appear together in the study of logical depth and resource-bounded complexity. These notions warn us that ‘little information’ does not mean ‘little work’. They depend on technical conventions and parameters; we will not use a label such as ‘deep’ as an intuitive score of an idea’s value. [6]

6.4. What we can measure without claiming the impossible

The length of the shortest program for an object is a useful mathematical notion, but there is no general procedure to compute it for all objects. In practice, the length of the best description found is an upper bound, not a certification of the absolute minimum. The general limits of computability make this modesty necessary. [5]

We can measure exactly the size of a model serialized in a fixed format. We can measure time on a declared platform, peak memory and the number of primitive operations. We can estimate energy consumption only with an appropriate method; the operation count is not automatically an energy measurement. We can report search cost and the latency distribution, not merely a convenient average.

More importantly, we can construct lower bounds for restricted classes. If a task requires distinguishing m situations that necessarily have different continuations, an exact deterministic representation needs at least m distinct states. This is not a bound for every probabilistic system or every approximate task, but it is a precise claim that can guide an experiment.

6.5. The cost–fidelity frontier

We will not necessarily seek a single winner. One model may save memory and consume more time; another may respond quickly but lose fidelity under interventions. We represent each candidate by a vector of costs and errors. A candidate is dominated if another is no worse on any relevant criterion and better on at least one.

The nondominated models form a frontier of tradeoffs. In a concrete study we will use the term ‘cost–fidelity frontier’, not to introduce a new theory through a name, but to keep the plurality of objectives visible. Choosing a point on the frontier depends on the application and acceptable risk.

The human author’s question thus takes a more productive form: not ‘how many rules does the world have?’ but ‘what minimum cost do we know for preserving these consequences at this horizon and under

these conditions?' An answer can be local yet profound. It shows which part of complexity belongs to the mechanism and which belongs to the representation or the observer's requirement.

7. The past worth preserving

7.1. Two histories, the same possible future

Imagine a device that counts button presses modulo three. Histories with 3, 6 or 300 presses may differ in length but be equivalent for continuation if the device retains no other information. To predict the next signal, the remainder on division by three matters, not the list of all presses.

The generalization is natural. Two histories are predictively equivalent if, under the permitted conditions, they induce the same distribution over all relevant future observations. In the version without actions, we can write:

$$h_1 \sim h_2 \Leftrightarrow P(\text{future} \mid h_1) = P(\text{future} \mid h_2)$$

This is the idea underlying causal states in computational mechanics. The theory characterizes minimal predictive representations under its conditions, not the metaphysical identity of the hidden mechanism. The term ‘causal’ in this tradition does not automatically mean identifying intervention effects in the sense of structural causal models. [9]

For a researcher, the definition’s value is that it replaces superficial resemblance with a functional criterion. Two texts may differ lexically and be identical for a particular continuation task. Two almost identical histories may require different states if one detail changes what follows.

7.2. Why a larger window does not always solve the problem

Suppose the device emits the same symbol at each press and a special signal only when asked about the remainder. A window seeing the last ten presses cannot generally determine the total remainder. Lengthening the window postpones the problem but does not eliminate it for arbitrarily long histories. A three-value counter solves it through recurrent updating.

This example does not show that every finite-context model is useless. It isolates a class of dependencies for which the correct representation is an accumulated state, not a longer suffix. In other tasks, local context is exactly what is needed. The program should discover the difference rather than declare one architecture universally superior.

There is another contrast. A process generated by a few hidden states does not guarantee that the observer needs only a few predictive states. When observations do not reveal the hidden state, the observer may maintain a belief distribution over it. That distribution can take very many values, even a continuum. The complexity of the generator and the complexity of inference about it can diverge radically.

7.3. A compression must be able to continue

If two histories are grouped into the same state, the representation must update them compatibly after the same new observation. Otherwise, the grouping does not define a well-determined recurrent mechanism. This is the requirement that the state be closed under updating.

A summary can be excellent for one question and useless for the next. ‘The temperature is acceptable’ may answer an immediate check but does not preserve the distance to the threshold. After slight heating, we do not know whether to change the conclusion. In a system that must operate continuously, the summary must also retain the information needed for updating.

This yields an important technical hypothesis: some forms of understanding can be characterized by finding a summary that is simultaneously sufficient for questions and closed under the relevant transformations. This is an operational proposal, not an exhaustive definition of human understanding.

7.4. Approximation as selection of useful information

Exact equality of future distributions is often too demanding. We can tolerate some predictive loss to reduce memory. The information bottleneck method formalizes compression of one variable while preserving information about another. In our problem, the first variable can be the past and the second the future of interest. [10]

A working formulation seeks a representation Z that minimizes the information retained about the past, subject to a bound on predictive distortion:

minimize $I(\text{past}; Z)$, with predictive distortion $\leq \delta$

I is mutual information, measured in bits when using the base-two logarithm. It is not the total implementation size. Distortion can be defined through a difference between future distributions or through loss on the relevant task. Theories of predictive inference and causal rate-distortion provide foundations for this choice. [11, 12]

The proposed program extends the practical question: an informationally compact representation may be very expensive to compute or update. We must therefore measure memory, discovery, updating and interventional fidelity separately. Not every elegant solution to the informational problem is a good engineering solution.

7.5. When ‘almost the same’ does not define classes

If the distance between two distributions is at most δ , we can say they are close at that tolerance. But closeness is not transitive. The first distribution may be close to the second, the second to the third, and the first too far from the third. A chain of local merges can produce a class with a large diameter.

This observation has direct consequences for learning. We cannot successively merge states merely because each merge seems acceptable at the time. We must control the error of the entire class, propagation through transitions and the effect on future queries. Otherwise, compaction may manufacture a certainty that the data do not support.

A valuable result would be a procedure that proposes merges, identifies observations that could refute them, and breaks up the grouping when new evidence appears. Such a memory does not grow only by addition or shrink only by forgetting. It changes its organization in response to consequences.

8. Equivalence has an observer and a budget

8.1. Several questions hidden in a single word

Are two mechanisms identical if they have the same structure? If they emit the same strings? If they assign the same probabilities to strings? If they respond alike when we modify them? If no available test can distinguish them? Each question defines a different relation.

Input-output equivalence can ignore internal details. Bisimulation requires a correspondence between transitions and states in a more structural sense. Simulation can be directional: one system reproduces another's behaviors without the converse being true. Bisimulation metrics for decision processes show how differences with consequences for evaluating decisions can be measured. [13]

We do not need a relation to replace them all. We need to choose the relation appropriate to the claim. Compressing an observation log must not be presented as discovering the mechanism. A causal correspondence must not be confused with the physical identity of implementations.

8.2. A common contract for two models

We fix a common observation space, comparable initial conditions, a family Π of intervention policies and a horizon H . A policy may choose the next action based on observations already received. We denote by $P(M, \pi, H)$ the distribution of the entire observed trajectory in model M under policy π .

$$d(M, N) = \sup\{TV(P(M, \pi, H), P(N, \pi, H)) : \pi \in \Pi\}$$

Total variation distance, TV , is the largest difference in probability that the two distributions assign to the same observable event. For finite discrete spaces, it is half the sum of the absolute differences between

probabilities. A value of zero means distributional identity under the fixed contract; a value of one permits perfect separation through an event.

Strictly speaking, this is generally a pseudometric on models: different internal mechanisms may have zero distance. It can become a metric on classes identifying models indistinguishable under the chosen contract. The word ‘distance’ must not be interpreted as a measure of the ontological difference between mechanisms.

Before using the formula, the models’ observations and interventions must be aligned. If this alignment is chosen differently for every pair, properties such as the triangle inequality are no longer automatic. For an initial theory, fixing a common interface is safer than simultaneously optimizing the meaning of observations.

8.3. A finite observer changes the problem

TV allows, by definition, any test on the trajectory. A real observer has limited time, memory and samples. We can restrict the comparison to a class D of tests that respect a declared budget:

$$d(D; M, N) = \sup\{ |P(M; T=1) - P(N; T=1)| : T \in D \}$$

This formulation has a precise relative in cryptography: computational indistinguishability. Under hardness assumptions, distributions generated from short seeds can be difficult for computationally limited observers to distinguish from uniform randomness. The existence of general constructions is conditional on assumptions such as the existence of one-way functions; it is not unconditional proof that every apparently random source hides a short rule. [28]

An analytical example shows the separation. A deterministic generator transforms an m -bit seed into an n -bit string, where n is greater than m . The output support contains at most 2^m elements. The uniform distribution over all n -bit strings assigns this support probability at most 2^{m-n} . The statistical distance from uniform is therefore at least $1 - 2^{m-n}$. Yet efficiently identifying the support may be difficult.

The conclusion is not that reality is pseudorandom. It is that generator simplicity, statistical distance and the observer's difficulty are different axes. A program about worlds and representations must keep them separate.

8.4. What a test actually measures

A finite collection of tests can discover a difference. If a test separates the models, we know they are not equivalent in the class containing it. If all attempted tests fail, we do not automatically know that no other permitted test would succeed. Conceptually, the observed maximum is a lower bound on the supremum over all tests, not an upper bound.

Strong positive claims require either fully enumerated test classes, mathematical guarantees, or statistical bounds for a controlled class. In large investigations, accurate reporting may be more modest: 'we identified no differences through these tests, with this detection power'. Precise modesty is more informative than a general declaration of equivalence.

Also, expanding the class of observations, policies or the horizon cannot decrease a distance defined through a supremum, provided the interfaces remain compatible. More opportunities for investigation may reveal hidden differences. This elementary monotonicity provides a consistency check for metrics implemented in a study.

8.5. A family of frontiers, not a scale of existence

We can seek models that respect a tolerance ϵ and minimize costs. But the resulting frontier depends on H , Π , observations and the test budget. A cheap model for immediate forecasts may become expensive when we demand long-term control. A compact representation for a single observer may lose its advantage when it must serve multiple questions.

These frontiers do not classify entities as 'more real' or 'less real'. They classify the adequacy of representations. The philosophical benefit is precisely avoiding a confusion: a world's resistance to our descriptions can be investigated without turning a model's success into a definition of existence.

9. What observation does not decide: intervention

9.1. The same data, two mechanisms

Consider two lights, A and B, and a hidden switch U that takes the values zero and one with equal probabilities. In the first mechanism, A is controlled by U and B copies A's state. In the second, both lights are separately controlled by U. Looking only at the lights, we always see the same state in both.

The structural descriptions are $A=U$, $B=A$ for the first mechanism and $A=U$, $B=U$ for the second. The observational distributions coincide. Yet if we force A to stay off, replacing the mechanism controlling it, the first system also turns B off; in the second, B continues to follow the hidden switch.

This is a constructed example, not an experimental finding about a device. It proves by enumeration that the same observational distribution can be compatible with different interventional effects. No additional amount of identical passive observations can decide between these two descriptions without additional assumptions or access.

9.2. Conditioning does not mean intervening

'What happens when I observe $A=0$?' and 'what happens when I force $A=0$?' are different questions. In the first, the observation tells us something about A's causes. In the second, we modify the mechanism producing A and may break that link. Equality between the two operations requires conditions; it does not follow from notation.

In a learning system, actions may be chosen precisely in situations where the outcome was already likely. If we ignore selection, we attribute to the action an effect it did not produce. For the proposed experiments, randomizing actions in controlled domains is a simple way to separate these problems. In uncontrolled environments, identification assumptions must be documented.

A sequential predictor is not excluded from this research. It can learn distributions including interventions if the data, representation and assumptions permit this. The error would be to attribute to an architecture an inability to represent causality merely because a particular dataset does not identify it.

9.3. When a small world explains a large world

A causal abstraction links a detailed model to an aggregate one. A map τ transforms detailed states into abstract states. A correspondence ω links interventions at the two levels. The intuitive requirement is that ‘intervene and then abstract’ should coincide, exactly or approximately, with ‘abstract and then intervene accordingly’.

$$\textit{intervention} \rightarrow \textit{abstraction} \approx \textit{abstraction} \rightarrow \textit{intervention}$$

The literature on causal consistency and approximate causal abstraction formalizes precisely such relationships between levels. It provides a foundation for discussing equivalences that preserve intervention effects, not merely correlations. [14, 15]

One example might be the transition from molecules to a pressure variable. An intervention called ‘increase pressure’ may have several microscopic realizations. If these realizations produce very different abstract outcomes, the high-level intervention is underspecified. A map between states is not enough; we must also understand what actions the map represents.

9.4. The cost of a causal distinction

A representation merging the two light mechanisms is perfect for passive observation but wrong for intervention on A. Preserving the distinction requires additional information. This suggests that the memory needed for control may exceed the memory needed for forecasting under a fixed regime.

This is not a universal law of parameter count. Sometimes causal structure simplifies the representation and reduces learning cost. The hypothesis worth testing is conditional: when the observation distri-

bution hides differences relevant to action, compression optimized exclusively for observations may eliminate information that control would need to preserve.

A good study would compare frontiers in two regimes: passive observation and intervention. If the frontier shifts, the reason must be identified. Perhaps new states have appeared, perhaps the model needs different relations, or perhaps a single additional variable suffices. What matters is the mechanism of the shift, not merely a worse score.

9.5. Experiment as an operation of separation

When two models explain the same data, the next question should not be ‘which seems more convincing?’ but ‘what observation or intervention would separate them?’ A research agent could seek actions maximizing predictive disagreement under cost and safety limits.

Such an agent does not discover truth through disagreement alone. The models may all be wrong or share the same hidden assumption. The program must therefore also include exploratory actions not justified exclusively by current models. The value of active experimentation lies in reducing specified ambiguities, not in guaranteeing an exhaustive space of explanations.

10. Irreducibility, chaos and legitimate shortcuts

10.1. Knowing the rule does not mean knowing the result immediately

A mechanism can be described in a few lines yet require many steps to produce a distant result. Wolfram's notion of computational irreducibility draws attention to the impossibility, in certain situations, of replacing unfolding with a general predictive shortcut. Broad claims about the phenomenon's prevalence must be separated from hardness proofs for precise classes and tasks. [7]

For our program, the correct question is always qualified: irreducible for which property, with what type of algorithm, at what precision and relative to which resources? Computing every cell of a configuration may be hard, while estimating a density or symmetry may be easy.

The existence of predictable aggregate descriptions for systems with difficult microscopic dynamics is explicitly investigated in the work of Israeli and Goldenfeld. It gives us reason not to confuse the complexity of detailed evolution with the impossibility of any useful large-scale theory. [8]

10.2. Three difficulties that must not be mixed

The first difficulty is informational: we do not know the initial state precisely enough. The second is computational: we know the relevant data, but computation is costly. The third is inferential: we do not know which mechanism in a class explains the observations. They may occur together but require different remedies.

Sensitivity to initial conditions shows that small state errors can produce large differences later. It does not by itself prove that fast algorithms do not exist for certain observables. Conversely, an exact discrete problem can be computationally difficult without a measurement error or a phenomenon of chaos in the usual sense.

In an investigation, we can separately vary initial precision, computation time and the amount of learning data. If performance improves only when precision increases, we have not demonstrated a limit of representation. If the correct model, supplied directly, remains slow, we have separated computational difficulty from learning difficulty.

10.3. From local error to trajectories

Suppose that for every relevant history two models have next-observation distributions at TV distance at most ϵ . We can couple the continuations so that the probability of a first divergence at each step, conditional on previous agreement, is at most ϵ . Over H steps, the probability of a divergence is at most $1 - (1 - \epsilon)^H$ to the power H .

$$d(H\text{-step trajectories}) \leq 1 - (1 - \epsilon)^H \leq \min(1, H\epsilon)$$

This is an elementary bound, not a guarantee provided by a small average log-loss. The assumption is uniform over relevant histories, including rare ones that may occur during free generation. A model can have an excellent average and a large error in an infrequent but decisive situation.

The bound explains why the fidelity of a long continuation must be evaluated separately. It does not prove that error must grow linearly; it is a general upper bound under the given conditions. The process's particular structure may yield better behavior or make the bound nearly relevant.

10.4. When the system dampens errors

For some dynamics, closeness between states may obey a relation of the form $\delta(t+1) \leq \rho\delta(t) + \eta$, where η is the error introduced at one step and ρ is less than one. The state error then remains controlled by a

geometric sum and need not grow without bound. This property must be proved in the chosen metric, not assumed for every system that ‘seems stable’.

$$\delta(t) \leq \rho^t \delta(0) + \eta(1-\rho^t)/(1-\rho), \text{ for } 0 \leq \rho < 1$$

Controlling final-state error is not identical to closeness between the distributions of entire trajectories. Two processes can reach similar final distributions after very different histories. If the task tracks every event, final contraction does not erase differences already observed.

This distinction suggests two types of approximation: some preserve step-by-step evolution; others preserve only final results or statistics. Both can be valuable, but they have different contracts. A legitimate shortcut must state which type it offers.

10.5. General limits do not close off research

There is no universal procedure that decides the equivalence of all general programs. A simple argument would turn such a procedure into a halting decider: compare a mechanism that constantly emits zero with one that emits zero until another execution terminates, then emits one. Identity of all outputs would decide whether that execution never halts. The limits established by computability theory prevent this generality. [5]

But for finite automata, finite horizons, controlled structural classes or restricted properties, we can obtain complete procedures, guaranteed approximations and certificates. A universal limit tells us where to delimit the problem, not that the delimited problem lacks value.

The program should therefore seek islands of verifiable equivalence and study their boundaries. When we leave an island, we may lose exactness, efficiency or the possibility of automatic discovery. Identifying what is lost, and why, would be a more durable contribution than declaring everything reducible or nothing capable of being shortened.

11. The extended family of next-symbol prediction

11.1. A common objective, different mechanisms

Any joint distribution over a finite string can be decomposed into successive conditional probabilities. For conditioning events with positive probability, the chain rule gives the identity:

$$P(x_1, \dots, x_n) = P(x_1) \cdot P(x_2 | x_1) \cdot \dots \cdot P(x_n | x_1, \dots, x_{n-1})$$

The identity does not specify how the probabilities are computed. They may come from a table, an automaton, a hidden-state model, a simulator or a neural network. Next-symbol prediction is a common objective, not an ontology or a unique architecture.

The link between prediction and compression is technical: probabilities assigned to sequences can be used to encode them, and predictive performance is reflected in description length. Research on language models as compression mechanisms develops this perspective. By itself, it does not establish whether the internal representation identifies the causal mechanisms of the world described. [20]

In an LLM with frozen weights, the procedure producing the next-token distribution remains fixed. Context and execution state change. If the weights are also updated, the parameter-state pair can be included in an extended dynamics. The novelty of generated sentences does not demonstrate the emergence of new fundamental instructions in the inference mechanism.

11.2. Memory through contexts: the nearest neighbor

A fixed-order model estimates the next symbol from the last k symbols. It is easy to interpret: each context has a distribution of continuations. The disadvantage is that the number of possible contexts can grow rapidly with k , while a recurrent dependency may remain hidden beyond any fixed window.

Variable-context models allow some situations to use a short past and others a longer one. Context-tree weighting combines models organized in a tree and has compression guarantees for the source classes and conventions analyzed in its theory. These guarantees do not mean that every semantic dependency will be learned efficiently. [17]

This family is an indispensable benchmark. Before proposing sophisticated mechanisms for ‘understanding’, we must see how much can be achieved through statistical reuse of contexts. An improvement over a memoryless predictor is not enough; the relevant comparator must benefit from the memory the problem permits.

11.3. Memory through state: different histories, the same role

A probabilistic automaton can have an internal state updated at each observation. Unlike a suffix tree, it can bring very different histories into the same state. This can allow compact representation of recurrent dependencies, such as modular counting or alternating phases.

The difficult problem is not merely estimating probabilities, but finding the state structure. Algorithms exist for discovering sequential patterns that construct hidden states from data and provide convergence results under explicit assumptions. They are a starting point for a new program, not a historical detail to ignore. [18]

A study should compare two ways of increasing memory: adding contexts and discovering a recurrent state. When an advantage appears, the important question is whether it comes from a better partition of

histories, a structural assumption supplied by the designer, or a budget difference. These three explanations have different scientific values.

11.4. Hidden regimes, counters and stacks

Hidden-state models separate the generator's state from visible observations. Filtering updates a belief distribution over the state. Regime-change models can track changes in local regularity; Bayesian online changepoint detection is a formal approach to this problem. [19]

A counter adds a form of memory that grows numerically. A stack allows nested structures to be managed: an opening is retained until its corresponding closing appears. For a family of sequences with nesting without a fixed limit, a stack expresses a regularity that a finite collection of contexts cannot represent exactly at all lengths.

It does not follow that stacks are the general solution for language. They provide a laboratory in which we know which dependency we seek and can verify transfer to unseen depths. A model may learn the surface format yet fail at deeper nesting. Another may receive its stack from the designer; its success verifies the structure's usefulness, not its autonomous discovery.

11.5. Predicting experiments, not only symbols

Predictive State Representations propose describing state through predictions about the outcomes of sequences of actions and observations. Instead of directly asserting which hidden variable exists, we retain a set of predictions that can be updated and used to anticipate other tests. [16]

This is a particularly fruitful relative of next-token prediction. The question becomes 'what will I observe if I perform this sequence of actions?' rather than merely 'what appears if I keep watching?' For the research program, this perspective naturally connects memory, experimentation and control.

Nevertheless, it must not automatically be called a complete causal theory. Data-collection policies, identifiability and action coverage remain essential. A predictive representation may be correct only within the domain in which it was defined and trained. Expanding the actions may require new internal distinctions.

11.6. Why good prediction does not certify rule compliance

A model may assign a very small probability to a forbidden continuation. If that probability is positive, repeated generation may produce a violation. Conversely, an exactly imposed constraint may eliminate the continuation but may be wrong if the real world allows exceptions.

Distribution quality, fidelity of generated sequences and validity guarantees must be separated. Regularization or smoothing may improve statistical estimation and avoid premature zero probabilities, but they do not turn an event's absence into logical impossibility. A system that must respect an invariant needs an additional validation mechanism or a generative class that preserves it by construction.

This does not establish a simple hierarchy between symbolic and neural approaches. A symbolic mechanism can contain incorrect rules; a neural model can learn a useful structure. The question is which part is estimated, which is imposed, which is verified, and what happens when these parts conflict.

12. From unfoldings to reusable operations

12.1. Two forms of compression that meet

We can compress the past into a state. We can also compress a repeated calculation into an operation. The first compression answers ‘what must I remember?’ The second answers ‘what work can I avoid repeating in the same form?’ The human author sees in their meeting a possible partial explanation for growing competence.

Suppose an agent learns to repeatedly compute the effect of several transformations. At first it applies every step. Later it discovers that a particular composition has a simpler form valid for a class of inputs. If it preserves that form and its validity conditions, its library grows and the time needed for some tasks decreases.

The novelty is real at the level of the agent’s repertoire without requiring new fundamental laws. But library growth is not automatically progress. A thousand procedures each memorizing one case can cost more than a single general rule. Transfer must be measured on inputs and compositions not used when inventing the procedure.

12.2. The circuit as a representation of dependencies

A computational circuit can be viewed as a network of values and transformations: a value is obtained from other values through a specified operation. This perspective makes the provenance of results visible. If a result depends on an observation, the link can be traced. If a subcomputation is reused, it must not be mistaken for new independent evidence.

For processes with memory, we can unfold the circuit through time or explicitly represent the state and its update. A finite circuit does not automatically capture every unbounded execution in finite size. A fixed horizon and a symbolic loop are different objects, even if they describe the same type of transition at different scales.

The advantage of research through circuits is not universality alone. Almost any sufficiently expressive formalism can represent many computations. The potential advantage lies in inspecting dependencies, separating observations from assumptions, local verification and replacing subcircuits with equivalent variants. These advantages must be demonstrated in applications, not assumed from a network drawing.

12.3. The contract of a new operation

An operation promoted into the library should have an input domain, an output meaning and a relationship to the procedure it replaces. The contract specifies whether equivalence is exact or approximate, which intermediate observations are preserved and which resources it saves.

A direct formula for a final state does not automatically replace an execution that must emit all intermediate states. Nor can it avoid the cost of writing a very long result when explicit output is required. An operation producing a compact representation of the result solves a different task from enumerating it, unless the interface accepts that representation.

The contract must also state what happens outside its domain. The operation may refuse, fall back to detailed computation or request additional information. Answering confidently outside demonstrated conditions is not an intelligent shortcut; it is a loss of the link between assertion and mechanism.

12.4. Learning libraries and the obligation to account for assistance

Learning programs with libraries of abstractions has important precedents. DreamCoder explores growing a library of reusable components together with mechanisms that guide search. It exemplifies the idea that learning can change the internal vocabulary of problems, not merely the parameters of a fixed vocabulary. [23]

For our program, the next question would be whether such operations can receive explicit predictive and interventional contracts. Compressing training tasks is not enough. We must verify that they do not destroy important distinctions and that the savings persist after including the costs of search, storage and validation.

An AI partner can propose candidate operations. In that case, its prior knowledge is a resource supplied to the system. We cannot describe success as discovery from local observations if the solution was suggested by a model trained on a vast corpus. We can nevertheless study separately the proposal's usefulness and the verifier's ability to reject incorrect proposals.

12.5. Two traditions of verification

Abstract interpretation develops semantic approximations used to analyze programs. A sound approximation may include all possible concrete behaviors and, sometimes, additional behaviors. Its purpose may be to prove the absence of an error, not to imitate observed frequencies. This notion of approximation is not identical to a small probabilistic distance. [21]

Formally verified compilation proves preservation of a semantics under the conditions of a specified language and implementation. Leroy's work provides a benchmark for what a real semantic obligation means in program transformation. It must not be extrapolated to certification of any summary or any model of reality. [22]

The program can borrow from both traditions: property preservation when a formal specification exists, and fidelity measurement when we have distributions and observations. When combining them, it

must keep the guarantees distinct. A proof about the model does not prove that the model corresponds to the world; a favorable test in the world does not certify every execution of the model.

12.6. A more ambitious technical objective

A mature system could retain provenance for each operation: which observations, rules or transformations it follows from; where it was verified; under what conditions it failed; and what cost it estimates. If a premise changes, dependent operations can be reevaluated. The library thus becomes a network of revisable commitments rather than a repository of claims without genealogy.

This is a proposed research architecture. We are not entitled to assume that provenance automatically resolves truth or that all dependencies can be correctly extracted from language. But we can construct bounded tasks in which genealogy is verifiable and measure whether it reduces contradictions, revision cost or error propagation.

13. Language, imagination and the hypothesis of a predictive mind

13.1. A sentence can promise more than it realizes

Let us assert that an object is indestructible, that a character knows the future, or that a machine instantly solves every problem. The statements are easy to produce. They do not supply the mechanisms that would realize the asserted properties. In fiction, this may be legitimate: the statement establishes the world's conventions. In a technical explanation, it leaves precisely the problem open.

The existence of a statement, a theory's consistency and a mechanism's realizability must therefore be separated. A theory may be coherent relative to certain axioms without being a physical description. A specification may be clear without having an efficient realization. A system may produce plausible answers without maintaining a coherent model of all their consequences.

This difference does not diminish imagination. Imagination allows a possibility to be explored before its realization is known. The problem arises when freedom to formulate is treated as proof of possibility, or when a story's local coherence is confused with closure of all its consequences.

13.2. Meaning as a family of commitments

A partial operational definition of meaning might concern the distinctions an expression imposes on observations and actions. 'The key opens the door' is not merely a proximity between words; in a test environment, it anticipates some outcomes of using the key and excludes others. Nevertheless, human meaning includes social conventions, intentions and experiences that this definition does not exhaust.

For research, we can restrict the domain to controlled interactive worlds. A text describes objects, relations and transformations. The system must answer questions and maintain the consequences of actions. We thus evaluate whether the representation preserves the description's operational commitments without claiming to have solved the entire philosophy of meaning.

The distinction between linguistic form and meaning is explicitly debated in Bender and Koller's work. It must be read as a theoretical argument in a debate, not as a general theorem definitively deciding the capabilities of all language models regardless of data, interaction or architecture. [27]

13.3. Introducing a rule into a story

In a textual world, the author can introduce the sentence 'from now on, blue objects float'. For the reader, this is a new rule. For the mechanism interpreting the story, it may be merely the addition of a condition to an existing structure. For a simulator lacking the concept of color, it may require extending the representation. For the physical world, the sentence changes nothing automatically.

The same statement thus crosses four levels: the act of formulation, the state of the representation, the interpretive mechanism and the possible referent. Many confusions about 'worlds with dynamic rules' come from mixing these levels. The program should measure where the modification occurs and what resources are needed to integrate it coherently.

An interesting test would introduce new rules that interact with old ones. The system must identify conflicts, priorities and indirect consequences. If it answers only examples repeating the wording, it has not demonstrated composition. If it handles unseen combinations in the defined domain, we have a signal of semantic reuse, not yet proof of general understanding.

13.4. The biological analogy and its limit

Prediction is a fruitful principle in modeling some perceptual processes. Rao and Ballard's predictive coding model proposes interactions between predictions and errors to explain properties of visual processing. This type of result motivates studying predictive mechanisms but does not identify all mental activity with next-word prediction. [25, 31]

The book's functional hypothesis is narrower. Some competencies might involve learning states that summarize experience and operations that shorten deliberation. An artificial system could test the usefulness of this combination. Even if successful, the result would demonstrate the mechanism's possibility, not that humans use it in the same biological form.

Supporting a claim about mind would require distinctive behavioral or neurobiological predictions. If two artificial mechanisms produce the same behavior, mere behavioral fit does not decide which mechanism is realized in the brain. The link between levels must be investigated, not decreed through analogy.

13.5. What remains outside the operational definition

Phenomenal consciousness, the value of an experience and the existential meaning of understanding are not resolved by a predictive metric. A system can have a useful representation without the program deciding whether it 'feels' anything. Likewise, a person may consider that they have understood a moral situation in a sense not reducible to anticipating the next event.

This limit does not weaken the technical project. On the contrary, it prevents it from colonizing problems for which it does not yet offer tools. We can study a component of understanding while explaining what it covers: preservation of relevant consequences under transformations and finite costs. What it does not cover must remain explicitly open.

The AI partner accepts this hypothesis as an investigable functional direction. It does not turn it into an identity between mind and algorithm or between meaning and compression. The human author may continue to uphold a broader intuition, but evaluating it must rest on additional arguments, not on the success of bounded models.

14. Discrete worlds in which we know what we are looking for

14.1. Why begin with constructed worlds

Simulating a constructed world provides access to the generating mechanism, hidden variables and distribution of initial conditions. The researcher can know the operational truth even if the agent sees it only partially. This asymmetry allows observation error, representation error and learning error to be separated.

The risk is choosing worlds that favor precisely the proposed method. To control it, the families must include favorable cases, unfavorable cases and controls lacking the structure sought. Parameter distributions and success criteria are set before final evaluation. A spectacular demonstration on a single mechanism is not a map of the domain.

The first laboratories should not begin with general natural language. There, interpretation, knowledge and reasoning errors are difficult to separate. A small alphabet and a transparent rule can reveal exactly why a representation works. Linguistic complexity will be introduced later, once we know which property we are tracking.

14.2. The recurrent-memory laboratory

The basic generator could be a modulo- m counter with increment, reset and query actions. The observer sees the actions performed and the responses, but not the internal value. We can vary m , episode length, reset frequency and query rarity. In one version all actions are visible; in another, some are hidden or corrupted by noise.

The tasks would be predicting the next response, generating a valid sequence, and answering after a planned sequence of actions. Comparators would include a memoryless predictor, fixed and variable contexts, a learned automaton and a model supplied with the counter structure. The last is a benchmark for performance when the correct representation is supplied, not a competitor in autonomous discovery.

The favorable hypothesis is that discovering a recurrent state enables transfer to much longer episodes without proportional memory growth. The alternative hypothesis is that this discovery is so costly or fragile that a context model remains more advantageous over practical ranges. Both results would say something important.

One control would replace the modular response with a random function of histories, without a compact rule supplied to the model class. If the method continues reporting ‘meaningful’ states, we must check whether it is inventing structure. Another control would arbitrarily rename symbols and permute representations to detect dependence on convenient conventions.

Relevant measurements would include predictive loss, number of rule violations, total memory, update time and learning cost. Transfer must be tested both at new lengths and at parameter values defining a new family. Success at greater lengths with the same modulus does not prove the agent has learned the general concept of modular arithmetic.

14.3. The nested-structure laboratory

A second family would use openings and closings of different types with a clear nesting rule. The task is not simple counting: a closing must correspond to the most recent unclosed opening. The observer may see the complete string or a version with missing elements. We can control depth, length and type distribution.

The decisive comparison would separate a supplied stack from a discovered one and from an approximate context-based solution. A favorable result would be stable generalization to unseen depths accompanied by identification of the memory mechanism. A more modest but useful result would be error reduction only over a finite interval, with a clear degradation frontier.

Nearly valid sequences containing a single rare violation should be included. The system must distinguish noise from a rule change. If every exception is immediately absorbed through a new rule, the model may lose generalization power. If every exception is rejected, the model may become unable to learn a new convention. This tradeoff offers a concrete measure of rigidity and plasticity.

A composition test would combine the counter and stack in one world without using this combination during training. Success would indicate component reuse in the defined domain. Failure could show that learned operations remain tied to their original context and have not yet become independent abstractions.

14.4. The laboratory of rules that seem to change

The third family would contain distinct regimes: for example, the same input is copied in one regime, inverted in another and delayed in a third. The regime may change according to a process known to the researcher. The observer is not given the regime label but may receive more or less informative clues.

Two stages are necessary. First, the number and nature of the regimes are supplied, and the system need only infer the current regime. Then these are unknown, and the system must propose and validate the structure. Confusing the stages would turn a filtering success into an unjustified claim about inventing rules.

We can measure change-detection delay, false-alarm rate, memory retained about old regimes and performance when a regime returns. An agent that relearns every returning regime from scratch has not built a reusable library. An agent classifying every fluctuation as a new regime may memorize noise.

The intended result is not a demonstration that dynamic rules are ‘actually’ always fixed. That formal reduction already has its conditions. The investigation would show when an extended-state description can be efficiently found from observations and when inference cost makes it impractical.

14.5. What an informative failure would mean

If an automaton with the correct structure works but learning it fails, we have a discovery problem. If even the correct automaton does not predict well after some actions are hidden, there may be an information limit. If the model learns but occupies more memory than the comparator, the representation advantage has not become a system advantage.

Each laboratory must be able to produce such diagnoses. The objective is not to offer a series of victories but to identify which resource becomes critical. This map would allow suitable architectures to be chosen and prevent extrapolation from a simple regularity to promises about language or general intelligence.

15. Scales, incomplete observations and causal mechanisms

15.1. The description-scale laboratory

A family of cellular automata would allow microscopic trajectories to be compared with aggregate properties: symbol density, persistence of regions, movement of patterns or the distribution of component sizes. Local rules may be known while the observer receives only aggregate blocks or sparse samples over time.

Not every rule producing complicated images should be labeled ‘irreducible’. The study should compare costs for concrete tasks and use available theoretical results only within their domain. Coarse-graining cellular automata provides precedents for seeking large-scale descriptions simpler than detailed evolution. [8]

The favorable hypothesis is the existence of aggregate variables with sufficiently autonomous dynamics for useful predictions. The alternative hypothesis is that aggregation loses information that later reappears in observables. In that case, the large-scale model would need memory, additional variables or history-dependent noise.

A protocol would vary block size, the interval between observations and the prediction horizon. It would compare a memoryless local model, one with memory and detailed simulation. The gain must be reported for the same observable, not by comparing an approximate average with an exact trajectory the other method was required to produce.

15.2. When eliminating details creates memory

Suppose we observe only a moving object’s position, not its velocity. Two situations with the same position can have different continuations. A memoryless position model is insufficient; a short history

can help estimate velocity. Apparently, the rule at the observed level depends on the past, even though the extended dynamics may be simple.

This situation generalizes the human author's intuition. Sometimes 'more rules' are the price of eliminating variables. The observer compensates for hidden information with history-conditioned exceptions. A better representation might replace those exceptions with a latent variable, but identifying that variable is not guaranteed.

An experiment would systematically control hidden variables and measurement noise. It would track whether predictive memory grows as observation deteriorates and whether this growth can be explained by uncertainty about the eliminated variable. A negative result would matter: perhaps the lost information is no longer recoverable from the observed past.

15.3. Equation discovery as a test of relative simplicity

For continuous dynamical systems, we can start from known equations and give the agent only measurements. Sparse identification methods seek compact combinations of terms in a function library. SINDy is an important example, and identifying partial differential equations extends this direction to spatially distributed phenomena. [29, 30]

The sparsity assumption must be made visible. An equation may be short in natural coordinates and long after a nonlinear transformation of observations. If the library includes exactly the generator's terms, the problem is more favorable than when relevant terms are absent. Success must not be presented as assumption-free discovery of the laws of nature.

A suitable investigation would have three arms: known favorable coordinates, transformed coordinates and partial observation. We could vary precision, sampling frequency and noise. The important result would be not only the recovered equation but the stability of predictions and interventions in unseen regions of state space.

If several equations perform similarly on the available data, the study must preserve the ambiguity. A short equation may be preferred as a hypothesis, but the choice does not prove the mechanism's uniqueness. New initial conditions or perturbations may be needed to separate them.

15.4. The observational-doubles laboratory

We construct pairs of mechanisms with the same passive distribution and different effects of some interventions. The pair of lights is the elementary form. Richer families can vary the number of variables, hidden dependencies and available action types. The agent must choose which experiment would reduce ambiguity.

Comparisons would include random intervention selection, a policy based on disagreement among models and a policy supplied with the causal structure. The last provides a benchmark for the ideal experiment in the given class. Intervention cost and number of attempts must be included, not merely the model's final correctness.

A favorable result would be a reduction in the number of interventions needed to select a correct causal class. An unfavorable result would be repeatedly concentrating on interventions that separate current models but ignore the true mechanism because it is absent from the candidate class. Detecting this situation requires test mechanisms absent from the training family.

This laboratory could produce a precise link between compression and experimentation: more compact representations may facilitate searching for interventions, but may also eliminate the distinctions that made them necessary. The program must not assume the effect's sign in advance.

15.5. Approximations that preserve control

A large-scale model should be tested not only through forecasting but also through its use in choosing actions. If an approximation predicts well in a passive regime but recommends actions taking the system

into an unknown region, control error may become large. The distribution under the agent's policy is not the same as the original data distribution.

A protocol would evaluate closed-loop performance, respect for limits and frequency of returning to the detailed model. An approximation might be valuable precisely because it knows when it no longer holds. Detecting departure from the domain would become a measurable capability, not merely a warning added to documentation.

Not every application demands high-risk control. Initial tests can use synthetic environments without external consequences. Extension to real systems would require domain-specific validation, independently of the general framework's elegance.

16. Growing libraries and worlds described through language

16.1. The operation-invention laboratory

An environment could supply compositional tasks built from a small core of transformations. Some combinations appear frequently, others only at evaluation. The agent can propose derived operations and retain them in a library. The researcher tracks whether the library becomes a reusable vocabulary or a collection of memorized solutions.

A fixed library, a library growing only through memorization, and one permitting parameterized abstractions must be compared. An additional benchmark would receive the useful library from the outset. The difference between it and the agent discovering the library estimates the cost of building the repertoire, without confusing it with the usefulness of an already available repertoire.

Measures would include task length after rewriting, total amortized time, number of reuses on new tasks and the library's complete size. If an operation is used only once, notational savings may hide additional cost. If an abstraction is reused in new structural contexts, there is a stronger signal of generalization.

16.2. Lexical novelty, functional novelty, extension of primitives

The program must distinguish three events. The first is introducing a new name for an existing operation. The second is discovering a composition that solves tasks previously inaccessible within the available budget. The third is receiving an external primitive with a capability unavailable in the declared core.

The second event is particularly important. Although it does not increase a universal core's theoretical computability, it can increase the agent's practical power. Under a finite budget, what can be accomplished matters more than what could be computed given unlimited time. This is a precise form of resource-relative novelty.

The third event must be accounted for as an external contribution. If the agent receives a service solving a class of problems, it has not discovered that capability by itself. It can nevertheless learn to combine it efficiently. Designing an open system is not an error; hiding that openness during evaluation would be.

16.3. What open-ended growth can mean

A world with composite objects and interaction rules can produce increasingly varied structures. Research on open-ended evolution discusses conditions under which persistent novelty arises and the relevance of environmental architecture. This offers a vocabulary for the human author's question about dynamic repertoires without proving that every generator of variety has an unlimited capacity for innovation. [26]

In a finite study, 'we observed no saturation' does not mean 'there is no limit'. We can, however, define local indicators: the rate at which new capabilities appear, diversity of consequences, transfer between tasks and the cost of maintaining old competencies. The indicator must avoid rewarding mere accumulation of labels.

A strong test would require novelty to change a set of problems solvable under a fixed budget. An additional test would check whether the new operations can be explained through compositions of the core. If so, we have observed an extension of the practical repertoire, not a change in fundamental laws. If not, we must first check whether we have omitted an information source or external operation.

16.4. The coherent-textual-worlds laboratory

A test textual world can contain objects, places, possessions, events and conditional rules. The evaluator knows the reference semantic state. The same world can be described through different formula-

tions, while similar descriptions may hide differences of negation, time or condition. The agent sees the text and receives questions or actions.

The first stage would separate text interpretation from executing consequences. One arm receives the semantic representation directly; another must extract it. The difference between them locates the cost of language. A third arm may receive candidate interpretations and must preserve ambiguity until a question or observation resolves it.

Tasks would include updating an object's location, identifying a contradiction, answering a counterfactual question and integrating a new rule. Cases must be introduced in which the correct answer is 'this does not follow from the available information'. A model that always fills gaps may appear fluent but loses the distinction between fact, assumption and possibility.

The intended result would be preservation of consequences under paraphrase and composition, not similarity of the answer to a reference phrase. Different wording may be perfectly correct; nearly identical wording may reverse a negation. Semantic evaluation needs criteria independent of expressive style.

16.5. Revising beliefs without arbitrarily rewriting the world

At a more difficult stage, the agent receives information refuting an earlier assumption. It must change conclusions dependent on it and preserve independent conclusions. Claim provenance may help localize revision, but we must check whether dependencies were represented correctly.

We can measure the number of correctly retracted conclusions, unnecessary retractions, persistence of contradictions and recomputation cost. A model rewriting all conclusions may be correct but inefficient. A model preserving too much may remain inconsistent. A library of operations could accelerate revision only if validity domains are explicit.

This laboratory approaches the intuition of a world that must continue consistently. It does not assume that every story has a single interpretation. On the contrary, it can explicitly treat multiple worlds compatible with the text. The requirement is that the agent not silently change the reference world merely to produce a convenient answer.

17. How we recognize a result and avoid self-deception

17.1. Each claim has its own measurement

‘Predicts well’ may mean low log-loss, calibrated probabilities or frequently choosing the correct symbol. ‘Generates coherently’ may mean respecting invariants over long sequences. ‘Understands interventions’ may mean correctly estimating action effects. These properties must be evaluated separately.

Log-loss penalizes low probability assigned to the realized event. Calibration compares observed frequencies with stated probabilities in groups of predictions. Choice accuracy can ignore the complete distribution. None automatically replaces the others, and aggregating them into a single score may hide important failures.

For constraints, we should report violation frequency as well as time to first violation, severity and context. For long-term predictions, evaluation occurs at multiple horizons. For adaptive systems, the recovery period after a change must also be measured, not merely performance after stabilization.

17.2. Absence of violations is not a proof

If n independent trials, with the same unknown failure probability p , produce no failure, the probability of that outcome is $(1-p)$ to the power n . An exact one-sided upper bound at confidence level $1-\alpha$ is $1-\alpha$ to the power $1/n$. For $\alpha=0.05$ and large n , the approximation is close to $3/n$.

$p(\text{upper}) = 1 - \alpha^{1/n}$, after zero failures in n independent trials

This is a statistical bound under the stated assumptions, not a guarantee that p is zero. If the trials are strongly dependent, their raw count cannot be entered into the formula as though they were independent. A million symbols from a single trajectory do not automatically equal a million independent experiments.

For rare events, we can construct targeted tests that make them accessible. But failure frequency under the targeted distribution must not be reported as natural frequency without correction. There are two questions: how often we reach the critical situation and how well we behave once there.

17.3. Separating data and families

Training, model selection and final evaluation must use distinct information. If we repeatedly inspect the final set and change the method, that set has become part of development. To support transfer, we need held-out mechanisms, parameters or combinations, not merely other sequences produced by the same instance.

A protocol may have three levels: new instances from the same distribution, new parameter ranges and new mechanism families. Each level justifies a different claim. Generalization at the first level does not automatically imply generalization at the third.

Multiple random seeds help estimate variability but do not repair a conceptually narrow sample. A hundred runs of the same favorable world do not replace structural controls. Both variation between runs and variation between types of worlds must be reported.

17.4. Fair comparisons and real contributions

Comparators must receive comparable access to observations and interventions. If a method receives the number of states, a semantic schema or a primitive library, these advantages must be declared. We may include arms with additional information for diagnosis, but we do not mix them into a single ranking.

Total cost includes hyperparameter search, representation construction, data transformation, validation and response. For a pretrained model, marginal usage cost may be relevant to an application but must not be confused with the cost of discovering the capability from scratch. These are two legitimate perspectives answering different questions.

An ablation analysis would disable state merging, library growth, contract verification and active exploration in turn. If performance does not change, the component may be unnecessary in the tested domain. If removal improves results, we have evidence that the added mechanism introduces cost or error.

17.5. What an empirical estimate of equivalence cannot say

In very large spaces, TV distance between entire distributions cannot be estimated arbitrarily precisely from few observations without additional assumptions. In practice, we can evaluate selected events, discriminator classes or statistics of interest. These define useful tests, but weaker ones than exhaustive verification.

The report must specify where a guarantee exists and where there is only an attempt at detection. A proof for a finite class, complete enumeration over a horizon and a statistical test on data are different forms of support. We will not call all of them ‘validation’ without qualification.

Conversely, a failed search does not prove that a compact representation does not exist. Perhaps the algorithm did not find it, the language does not express it efficiently, or the budget was insufficient. An impossibility result requires proof within the declared class. A practical limit can be reported as a practical limit and remain valuable.

17.6. A register of claims

For every conclusion, a register should preserve the exact statement, assumptions, evidence and type of test that could refute it. A claim about memory savings must show what was counted. A claim about discovering a rule must distinguish what was supplied from what was inferred. A claim about transfer must identify what was new.

The register is not bureaucracy added to research. It prevents unnoticed migration of conclusions: from ‘works in this family’ to ‘explains understanding’, from ‘we found no violations’ to ‘it is safe’, or from ‘we compressed the description’ to ‘we discovered the world’s laws’.

A well-localized negative result can become more useful than a vague positive result. It reveals a missing variable, an incorrect assumption or a resource frontier. Precisely this information allows a future researcher to continue without repeating the same confusion.

18. Hypotheses that can lose and propositions that can be proved

18.1. The hypothesis of a state that replaces history

The first empirical hypothesis states that in families with identifiable recurrent dependencies, discovering an updatable state can favorably shift the cost–fidelity frontier compared with expanding contexts. The claim must be tied to concrete families, budgets and comparators. It does not say that the learned state will always be smaller or that every process has a finite representation.

An advantage persisting on longer histories and after including learning cost would support the hypothesis. An advantage disappearing when the comparator receives the same budget or when symbols are recoded would weaken it. Systematic failure under overly sparse observation, accompanied by proof that different hidden states remain indistinguishable, would delimit it.

The phenomenon sought is a regime change in representation. Initially, additional data may produce additional contexts. Later, a discovered regularity may allow them to merge into a common state. If such compression occurs, we must verify that it was not purchased by losing rare distinctions.

18.2. The hypothesis of compression's causal cost

The second hypothesis states that a representation optimized exclusively for passive observations may eliminate distinctions needed for interventions. In families with observational doubles, the difference can be demonstrated constructively. The empirical question is how often and how severely the effect appears in broader problem classes.

Interventional deterioration controlled by the degree of compression, while observational performance remains stable, would support the hypothesis. A contrary result would show that in the analyzed family the predictive structure already retains causal distinctions, or an additional criterion preserves them at low cost.

We are not seeking confirmation of the slogan ‘correlation is not causation’, but a quantitative map: which distinctions are lost, which interventions reveal them and what recovering them costs. Such a map could guide representation choices for agents that must act, not merely describe.

18.3. The amortized-shortcut hypothesis

The third hypothesis states that learning reusable operations produces a reuse threshold beyond which total cost decreases. If discovery and verification cost C , and each use saves s units compared with the reference procedure, a simplified accounting model has threshold C/s , with positive s .

$N \cdot s > C \Rightarrow$ total savings

This is an accounting calculation, not a theory of learning. In practice, s may vary across inputs, and the library adds selection and maintenance costs. The empirical hypothesis is that some tasks have a threshold that remains realistic after including these costs and that the advantage transfers to new uses.

One informative failure would be a proliferation of rarely reused operations. Another would be apparent savings obtained by changing the output format. The desired result is less work for the same contract, not merely a more elegant description of hidden work.

18.4. The hypothesis of memory's dual development

The fourth hypothesis states that an efficient learning system needs both expansion and reorganization. Expansion preserves new distinctions; reorganization finds regularities among them. A memory that only grows may become too costly. A memory that only compresses may lose important novelties.

The empirical prediction is the emergence of cycles of differentiation and reunification correlated with task changes. To avoid turning every fluctuation into confirmation, we must define in advance what constitutes a distinction, what constitutes a merge, and what minimum improvement justifies a change. The cost of structural instability must also be measured.

A negative result would be that repeated reorganization consumes more than it saves or causes systematic forgetting. In that case, a more rigid library segmented by domain may be preferable. The program does not assume that maximum plasticity is always optimal.

18.5. First proposition: closure of dynamic rules

Proposition. If state updates and rule updates are computable, and the necessary information is included in the extended state or explicitly modeled inputs, the system can be described by a single fixed computable function updating the extended state.

Justification. Consider the pair (x,r) . We evaluate x 's transition function using r and evaluate r 's update function using the specified data. The pair of results is the new extended state. The composition of these computations defines the fixed function G . If updates use simultaneous values or a particular order, that convention is included in G .

The proposition does not say that the state space is finite, that G can be deduced from observations, or that the resulting description is efficient. It is a structural clarification. Research begins when we ask what representing and discovering G costs in a concrete class.

18.6. Second proposition: composing representation errors

Proposition. If two successive transformations between distributions use compatible interfaces and have TV errors at most ε_1 and ε_2 in the declared domain, the composition error is at most $\varepsilon_1 + \varepsilon_2$. The result uses the fact that applying the same probabilistic transformation does not increase TV, together with the triangle inequality.

Justification. We introduce an ideal intermediate distribution. The first difference is the first approximation's error transported through the common transformation, so it cannot exceed ε_1 . The second is the second approximation's error, at most ε_2 . Their sum bounds both. The bound can be capped at one, since TV never exceeds one.

The result requires the second operation's guarantee to apply to the distribution it actually receives. If the first approximation sends the system outside the second's domain, the composition is not certified. An extension could add the probability of leaving the domain to the error budget. This offers a concrete direction for learned-operation contracts.

18.7. Third proposition: an exact state must separate incompatible continuations

Proposition. In an exact deterministic predictor, two histories admitting the same sequence of future actions but requiring different responses to it cannot be represented by the same internal state if updating and response depend only on that state and the actions.

Justification. Starting from the same state and applying the same action sequence, the deterministic predictor would produce the same response. This cannot coincide with two necessarily different responses. By contradiction, the internal states must differ.

This offers a method for constructing memory bounds: identify histories separable through continuations. Approximate probabilistic predictions require separation margins and additional criteria. We will not directly extrapolate the number of exact states to the minimum size of any neural or stochastic model.

18.8. An accessible theoretical question: what does separating candidates cost?

Assume a finite family of m models and a binary-outcome test for each relevant pair. Assume acceptance probabilities differ by at least Δ for the models we wish to separate and that we can repeat tests independently. Concentration of empirical means then allows a sufficient number of observations to be bounded, depending on Δ , the number of tests and the tolerated error level.

For example, for k fixed tests, the probability that any empirical mean deviates from its value by more than $\Delta/4$ is at most $2k \exp(-n\Delta^2/8)$, using a Hoeffding bound and the union bound. Therefore, $n \geq (8/\Delta^2) \ln(2k/\alpha)$ suffices for all these estimates to be simultaneously within tolerance with probability at least $1-\alpha$. [32]

This claim concerns test estimation, not automatic recovery of any model. Identification additionally requires that the tests actually separate candidates, that candidate-predicted probabilities be known sufficiently precisely, and that the relevant model be included in the family. Access to independent trials may be costly or impossible.

A research contribution could extend such results to active tests, varying intervention costs and libraries that change during learning. Instead of a universal promise of discovery, we would obtain verifiable conditions under which an ambiguity can be reduced.

19. Organizing research for humans and AI partners

19.1. Stages defined by evidence, not enthusiasm

The first stage would establish the operational vocabulary and verify analytical examples. The team must be able to say what a state is, what cost it measures and what type of equivalence it claims. Moving to the next stage requires agreement on these contracts, not agreement on the author's metaphysics.

The second stage would study discrete families. The aim is to locate mechanisms producing advantages or failures. A complex unified architecture is unnecessary. On the contrary, isolated components enable diagnosis. Further progress would require results robust to structural controls and complete resource accounting.

The third stage would add interventions, incomplete observations and scale variation. Here we check whether representations remain useful when the agent changes the distribution through its actions. Only after this stage would integration with richer textual worlds and libraries of AI-proposed operations be justified.

19.2. Which roles can be delegated

An AI partner can propose models, identify analogies with the literature, formulate edge cases and suggest interventions separating hypotheses. Another role is locally checking derivations and searching for counterexamples. These tasks are valuable precisely because they can be evaluated through verifiable products.

A system formulating a hypothesis should not be its sole evaluator of success. Multiple AI systems do not automatically constitute independent evaluators: they may share sources, assumptions and errors. Independence must be built through different methods, separate information access and verifiers that do not rely solely on verbal consensus.

Human researchers remain responsible for the question's meaning, the choice of relevant consequences and the use of conclusions. This responsibility does not disappear when computation or writing is delegated. Humans can also be biased; the process must therefore be designed for mutual criticism, not protection of the founding intuition.

19.3. A dossier for each important intellectual operation

A hypothesis should have a short dossier: why it was formulated, its connection to existing work, its premises, its distinctive prediction and the result that would prompt revision. An abstraction should have a contract. An empirical conclusion should have a protocol and a boundary on generalization.

Future researchers, human or AI, can use this dossier to continue the investigation without reconstructing initial intentions. Documentation must explain not only what was chosen but why alternatives were rejected. An unexplained failure invites repetition; a contextualized failure becomes reusable knowledge.

Conceptually, the dossier is itself an application of the program: it compresses a research history into a state sufficient for responsible continuation. It need not preserve every detail, but must preserve the premises and vulnerabilities that could change the next step.

19.4. Where an original contribution might emerge

Predictive state, rate-distortion, bisimulation, causal abstraction and library learning already exist as concepts. Simply bringing them together does not guarantee scientific novelty. An original contribution must be identified through a result: a new composition condition, an algorithm with proven properties, a reproducible diagnosis or a family of counterexamples that changes an assumption.

One promising direction would combine updatable predictive states with reusable operations having fidelity and cost contracts. Another would study how intervention choice changes the memory-prediction

frontier. A third would investigate local certificates allowing a system to switch from a fast approximation to a detailed mechanism when risk increases.

Causal abstraction is already used as a foundation for mechanistic interpretability of AI models. Any project linking internal states and operations to causal explanations must explicitly position itself relative to this literature. [24]

19.5. Stopping rules and continuation rules

A direction should be narrowed when its advantage disappears after resource accounting or fails to transfer beyond the examples that inspired it. Narrowing does not mean the general intuition has been refuted in all forms; it means a particular instantiation failed to support its promise.

A direction deserves continuation when it produces new, verifiable distinctions, even before surpassing every existing method. Discovering an observation limit, a condition for merging failure, or a separation between passive and interventional fidelity can justify a program of its own.

The criterion of maturity is not the framework's ability to rename any phenomenon. It is the ability to recommend a precise investigation, anticipate different results for rival hypotheses and turn an unexpected result into an explicit theoretical change.

20. What remains of the initial intuition

20.1. A world is not obliged to resemble our explanation

At the beginning, the human author's intuition opposes reality's consistency to representation's freedom. At the end of the analysis, the opposition remains fruitful, but not as an absolute classification. A text is a real process, fiction can be governed by strict rules, and a real phenomenon may remain difficult for its observer to compress.

What survives is the idea of an obligation to preserve consequences. An explanatory model cannot introduce the desired result for free without declaring a premise, operation or information source. A shortcut must show what it replaces. A new rule must state whether it is a definition, an inference, a convention or an actual modification of the mechanism.

This discipline does not prove that the universe has a small core of laws. It does, however, offer a method for avoiding confusion between simplicity of expression and simplicity of production. 'It is so' and 'it follows thus' may have the same grammatical form, but different obligations.

20.2. Understanding as the power to continue

A working formulation, accepted by the AI partner as an investigable hypothesis, is that a component of understanding consists in building a representation that can continue correctly under relevant questions and transformations using finite resources. This component involves selecting the state, preserving dependencies and recognizing the validity domain.

The formulation does not say that understanding is only prediction or only computation. It isolates a property we can test. A system repeating an explanation but unable to update the consequences of a simple change does not exhibit this property in that domain. A system exhibiting it is not thereby proven conscious or wise.

The human author proposes a broader intuition: perhaps what we call intelligence is partly the ability to construct internal worlds in which leaps become justified. Research must decompose this sentence into observable operations without losing its central question or turning its breadth into proof.

20.3. Approximation is not a shameful version of truth

An approximation is legitimate when it states which distinctions it no longer makes and why they do not affect the task under the declared conditions. Sometimes eliminating details is precisely what allows a regularity to be identified. At other times, the eliminated detail controls a decisive rare event. No general preference for simplicity alone decides between these situations.

The theory sought should therefore not offer a single final representation. It should allow navigation among representations: more detailed when new questions arise, more compact when repeating familiar tasks, more cautious when interventions leave the verified domain. Changing the level of description can itself be a competence.

Here approximation becomes a profound problem: what can we forget without losing the power to continue? The answer depends on the future we consider possible, the actions we permit and the costs we can bear. This is not arbitrary relativism but a precise relation among world, observer and purpose.

20.4. What we do not yet know

We do not generally know when an agent can efficiently discover the appropriate predictive state. We do not know how far automation of contracts for operations learned from language can be extended. We

do not know whether a unified framework will yield practical advantages large enough to justify its own cost. These uncertainties are research problems, not reasons to assert success or failure in advance.

Nor does the ontological question disappear. Perhaps the world has a compact generative description. Perhaps this compactness is local, level-dependent or insufficient for particular facts. An experimental program cannot settle all these possibilities alone, but can clarify their consequences for finite observers.

A useful theory might emerge from a convergence of small results: a recovered state, a refuted abstraction, a proven limit, a better-chosen intervention, an operation verified and reused. None would explain existence by itself. Together, they could better explain how we construct representations worthy of trust.

20.5. The question to leave future researchers

You need not begin by accepting that everything is computation or that every world reduces to a few rules. Choosing a system, an observer and a family of consequences is enough. Ask what information must be preserved, which transformations can be shortened and what result would demonstrate that you oversimplified.

If a smaller representation preserves more than you expect, seek the reason. If it fails on a rare detail, do not hide that detail in the average. If a new rule appears, ask where its cost lay before: in experience, search, the core of operations or an external source.

The human author's intuition does not ask to be protected from these questions. In its most valuable form, it demands them. An intelligible world is not one about which we can say anything, but one in which some passages between claims and consequences can be justified. This book's program begins where we try to discover what that justification costs and how much of it we can make explicit.

The working vocabulary

A.1. Terms that must remain distinct

Generative rule. A transformation determining a continuation or distribution of continuations under specified conditions. A rule is evaluated not only by the length of its name but by the description of the mechanism it invokes.

Metarule. A rule modifying other rules or their representations. If the modification is computable, it can be included in fixed dynamics of an extended state, without guaranteeing that the state is finite or easy to infer.

Derived rule. An operation defined by composing or transforming existing operations. It can extend the agent's practical repertoire without extending the class of functions computable by its core.

Generative state. The information used by the mechanism to determine the next transition. It need not be directly accessible to the observer.

Predictive state. The information the observer retains to anticipate continuations. It may be a distribution over generative states and may have a very different complexity from theirs.

Sufficiency. The property of a representation of preserving information needed for a declared class of questions. It does not mean preserving every property of the system.

Closure under updating. The ability to compute the new representation from the current representation and permitted inputs without silently recovering information purportedly eliminated.

Observational equivalence. Identity of observation distributions in the specified regime. It does not automatically imply identity of intervention effects.

Interventional equivalence. Agreement on observations resulting from a declared family of interventions, with a well-defined correspondence between the compared models.

Approximate closeness. A distance below a threshold. It does not automatically form equivalence classes because transitivity may fail.

Resource-bounded indistinguishability. The demonstrated impossibility or difficulty of separating two sources through a specified class of bounded tests. Failure of a few tests is not a proof for the entire class.

Abstraction contract. A description of the observations, interventions, domain and tolerance that a simplified representation commits to preserve.

Invariant. A property that must be preserved by all permitted evolutions of a model. Not having observed a violation does not, by itself, certify invariant status.

Functional novelty. An extension of capabilities achievable within a task family and a budget. Introducing new labels without new consequences is insufficient.

Cost-fidelity frontier. The set of nondominated tradeoffs between declared costs and errors. It depends on interface, horizon, task family and accounting method.

A.2. Notation

M denotes a model; S , the state space; A , actions; O , observations. K is the transition kernel, E the observation channel, and μ the initial distribution. The same symbol must not be reused for another quantity within the same protocol.

h denotes an observed history; Z , a representation of it; H , the prediction horizon. π is an action-selection policy, and Π a family of policies. D is a class of tests or discriminators.

L is a description length under the declared convention. I is mutual information. TV is total variation distance. ϵ and δ are error tolerances, while α is an allowed statistical error probability. They are not interchangeable: a model error and uncertainty about estimating that error are different things.

τ links detailed states to abstract states, and ω links interventions. C denotes a preparation or discovery cost, s a saving per use, and N the number of uses when the simplified amortization model is used.

The prospective investigation form

B.1. The question and the limited promise

The form begins with a question admitting different answers. For example: ‘In the chosen process family, does a learned recurrent state reduce total memory needed for a given predictive error compared with a variable-context model?’ Terms such as ‘real understanding’ or ‘emergent intelligence’ are insufficient as primary objectives without operationalization.

Next, formulate the favorable hypothesis, a rival hypothesis and a result that would prompt revision. If both directions of the result are immediately interpreted as supporting the same idea, the form is not yet ready for investigation.

B.2. The world and the observer’s access

Describe the generator, hidden variables, initial distribution, randomness source and conditions for regime change. Specify whether the agent sees all actions, can intervene and has access to resetting the environment. For a simulation, separate truth known to the evaluator from information supplied to the agent.

Declare the parameter family and the reason for choosing it. Record conventions that may favor a model: suggestive symbols, coordinates in which the equation is sparse, examples revealing structure or a known number of states. These can be deliberate, but must be visible.

B.3. Representations and resources

For each candidate, inventory the supplied mechanisms, learned parameters, structure that can grow and selection criterion. Describe how the state is updated and whether raw history is retained. Specify who can propose new operations and what external knowledge is available.

The budget includes preparation, search, storage, updating, response and verification. When a cost cannot be measured faithfully, declare it unknown or report it through an explicit proxy. Do not turn a proxy into a physical measurement without justification.

B.4. The contract and metrics

Fix relevant observations, permitted interventions, horizons and tolerances. Declare whether complete trajectories, final results, aggregate statistics or decisions are being compared. For abstractions, define the correspondence between interfaces before evaluation.

Assign a measure to each main claim. Prediction can be evaluated through log-loss and calibration. Coherence can be evaluated through specified properties. Control can be evaluated through action effects and costs. Transfer requires tasks differing in a declared way from those used for learning.

B.5. Data and controls

Separate development, model selection and final testing. Specify the independent evaluation unit: episode, generator instance or another justified unit. For temporally dependent processes, document the uncertainty-estimation method.

Controls include mechanisms lacking the regularity sought, symbol recoding, variation of information access and removal of method components. A benchmark supplied with the correct structure can separate representation difficulty from learning difficulty, but is labeled as such.

B.6. Interpretation before the result

The form specifies what each result class would mean. An advantage only in training suggests local adaptation, not transfer. An advantage only when structure is supplied shows the representation's usefulness, not its discovery. Loss of control after compression indicates a limit of the observational criterion. Higher total cost may cancel the latency advantage.

Also define a continuation rule: what new question would become justified if the hypothesis receives support? For example, moving from complete to partial observation, or from an isolated operation to unseen compositions. Continuation must increase difficulty in an interpretable way, not merely enlarge the demonstration.

B.7. Reporting for the next researcher

The final report should state directly what is proved, what is empirically supported, what is refuted within the studied domain and what has not yet been tested. Hypotheses changed after observing data are distinguished from those formulated beforehand. Relevant failure cases are preserved, not merely successful examples.

A good conclusion has clear reach. ‘In this family, at these budgets, the recurrent representation preserved these properties more cheaply’ is more valuable than ‘we created a system that understands’. The first can be checked and extended. The second can hide any result.

Reading map and bibliography

C.1. How existing traditions connect

For the philosophical question, useful entry points are the debate on laws of nature and the distinction between fundamental reduction and reconstruction of higher-level phenomena. [1, 2] These readings offer no mandatory common conclusion; they help recognize commitments hidden in expressions such as ‘law’ and ‘explanation’.

For simplicity and cost, the MDL principle should be read alongside computability limits and time-dependent notions. [3–6] This avoids both superficial counting of rules and the assumption that the shortest description is always the most useful.

For predictive memory, the recommended route goes from causal states and relevant information to rate–distortion and state-discovery algorithms. [9–12, 18] For action and comparison between levels, add predictive state representations, bisimulation and causal abstraction. [13–16]

For reusable operations, library learning, abstract interpretation and verified compilation should be brought into dialogue. [21–23] The first seeks new structures; the other two show what a precise semantic obligation means. The connection to AI interpretability can be investigated through causal abstraction. [24]

For scales and physical worlds, irreducibility, coarse-graining and equation discovery offer three complementary perspectives. [7, 8, 29, 30] For language and mind, readings on compression, meaning and predictive coding must remain distinct, without turning their analogy into a proven identity. [20, 25, 27, 31]

C.2. References

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C.3. The bibliography’s limit

This bibliography is a working foundation, not an exhaustive audit of the novelty of every proposal. Before claiming an original contribution, researchers must extend the search within the concrete subfield: system identification, automata learning, semantics, causal abstraction, complexity or interpretability. An interesting conceptual convergence does not negate the precedence of technical results.

When a source is cited as a manuscript, the identifier allows the relevant version to be found even if its title or subsequent publication date differs. The references do not certify the book’s prospective hypotheses. They identify the tools a rigorous program must understand before attempting to extend them.